

Labor markets and skills: labor mobility, skill-relatedness

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Economic Complexity, Class 5 – Oct 21 2025

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Recap and HW2: Urban scaling

$$Y(t) = Y_0 N(t)^\beta.$$

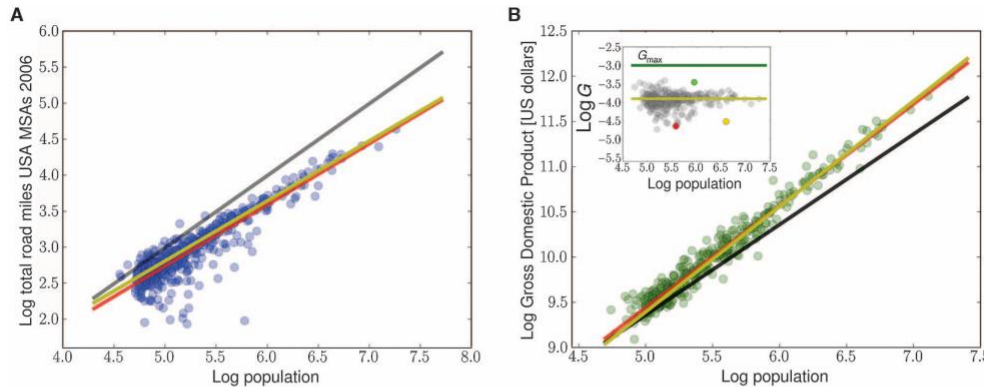


Fig. 1. Scaling of urban infrastructure and socioeconomic output. (A) Total lane miles (volume) of roads in U.S. metropolitan areas (MSAs) in 2006 (blue dots). Data for 415 urban areas were obtained from the Office of Highway Policy Information from the Federal Highway Administration (14). Lines show the best fit to a scaling relation $Y(N) = Y_0 N^\beta$ (red), with $\beta = 0.849 \pm 0.038$ [95% confidence interval (CI), $R^2 = 0.65$]; the theoretical prediction, $\beta = 5/6$ (yellow); and linear scaling $\beta = 1$ (black). **(B)** Gross metropolitan product of MSAs in 2006 (green dots). Data obtained for 363 MSAs from U.S. Bureau of Economic Analysis (14). Lines describe best fit (red) to data, $\beta = 1.126 \pm 0.023$ (95% CI, $R^2 = 0.96$); the theoretical prediction, $\beta = 7/6$ (yellow); and proportional scaling, $\beta = 1$ (black). The two best-fit parameters in each scaling

relation were estimated by means of ordinary least-squares minimization to the linear relation between logarithmically transformed variables (14). The inset shows the estimate of G for 313 U.S. MSAs and the conservation law $\frac{d \ln G}{d \ln N} = 0$ ($R^2 = 0.003$). G is measured as the product of gross domestic product and road volume, both per capita. As predicted by the theory, observed values of G for different cities cluster around its most likely value (mode, yellow line), which gives an estimate of the optimum G^* , and are bounded by the maximum $G_{max} \approx 8G^*$ (green line); see also Fig. 2B. Several metropolitan areas, such as Bridgeport, Connecticut (green circle); Riverside, California (yellow circle); or Brownsville, Texas (red circle), are outliers, suggesting that they are suboptimal in terms of their transportation efficiency or amount of social mixing.

Table 1. Scaling exponents for urban indicators vs. city size

Y	β	95% CI	Adj- R^2	Observations
New patents	1.27	[1.25, 1.29]	0.72	331
Inventors	1.25	[1.22, 1.27]	0.76	331
Private R&D employment	1.34	[1.29, 1.39]	0.92	266
"Supercreative" employment	1.15	[1.11, 1.18]	0.89	287
R&D establishments	1.19	[1.14, 1.22]	0.77	287
R&D employment	1.26	[1.18, 1.43]	0.93	295
Total wages	1.12	[1.09, 1.13]	0.96	361
Total bank deposits	1.08	[1.03, 1.11]	0.91	267
GDP	1.15	[1.06, 1.23]	0.96	295
GDP	1.26	[1.09, 1.46]	0.64	196
GDP	1.13	[1.03, 1.23]	0.94	37
Total electrical consumption	1.07	[1.03, 1.11]	0.88	392
New AIDS cases	1.23	[1.18, 1.29]	0.76	93
Serious crimes	1.16	[1.11, 1.18]	0.89	287
Total housing	1.00	[0.99, 1.01]	0.99	316
Total employment	1.01	[0.99, 1.02]	0.98	331
Household electrical consumption	1.00	[0.94, 1.06]	0.88	377
Household electrical consumption	1.05	[0.89, 1.22]	0.91	295
Household water consumption	1.01	[0.89, 1.11]	0.96	295
Gasoline stations	0.77	[0.74, 0.81]	0.93	318
Gasoline sales	0.79	[0.73, 0.80]	0.94	318
Length of electrical cables	0.87	[0.82, 0.92]	0.75	380
Road surface	0.83	[0.74, 0.92]	0.87	29

Productivity and the Density of Economic Activity (Ciccone & Hall, 1996)

- Research Question: Do areas with a higher spatial density of economic activity have higher productivity?
- Data: The authors used U.S. data from 1988. They combined state-level data on Gross State Product (as a measure of output) with more granular, county-level data on employment and land area.
- Method: The authors used an econometric model to link state-level productivity (Gross State Product) to a specially constructed index of county-level employment density. This allowed them to measure the impact of density itself, separate from other factors.
- Findings: Density has a strong, positive, and causal effect on productivity.
 - The main finding: Doubling employment density in an area increases average labour productivity by 6%. This effect is substantial and can explain more than half of the observed variation in productivity across U.S. states.

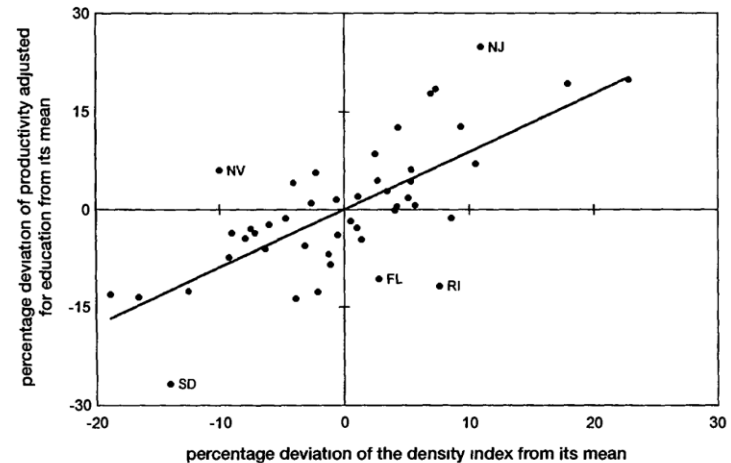
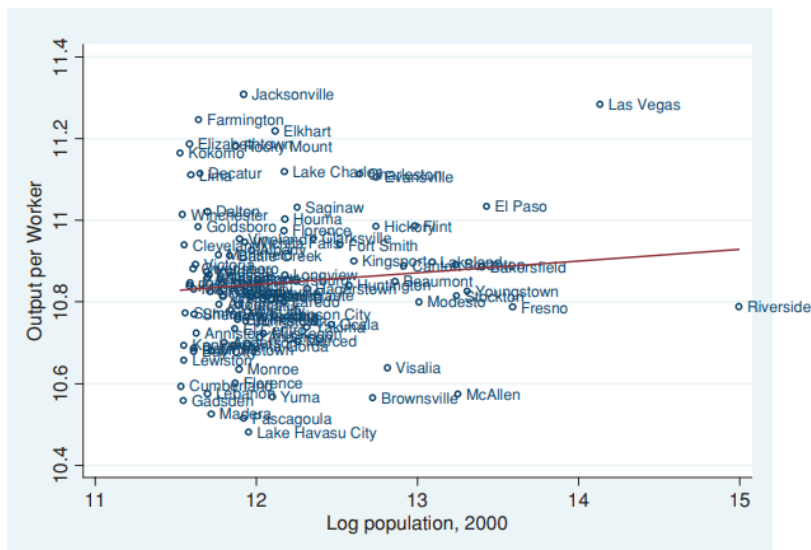


FIGURE 3. DENSITY AND PRODUCTIVITY ADJUSTED FOR EDUCATION, $\theta = 1.058$

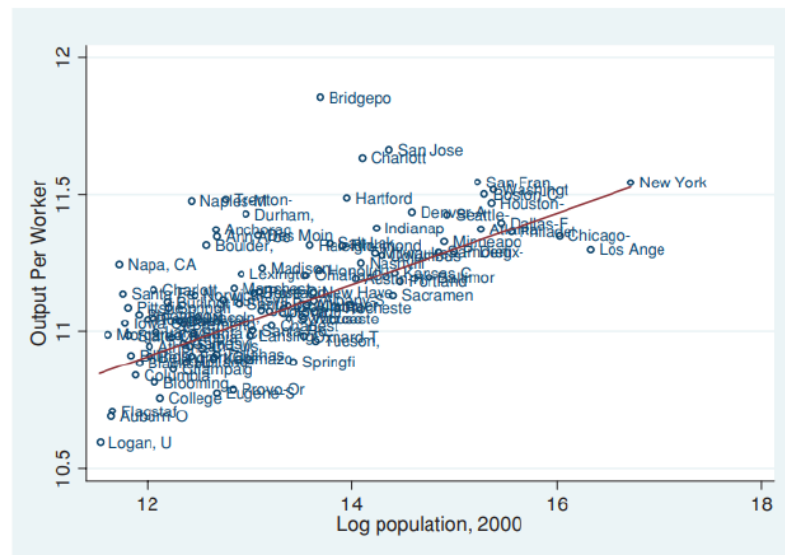
By Edward Glaeser

By Edward Glaeser



Notes: (i) Units of observation are MSAs under the 2006 definitions with populations above 100,000 and where the share of adults with college degrees is less than 17.65 percent. Labor force and population is from the Census, as described in the Appendix. Gross Metropolitan Product is from the Bureau of Economic Analysis.

(ii) The regression line is $\text{Log GMP per capita} = 0.028 [0.028] * \text{Log population} + 10.50 [0.34]$. $R^2 = 0.01$ and $N = 100$.



Notes: (i) Units of observation are MSAs under the 2006 definitions with populations above 100,000 and where the share of adults with college degrees is greater than 25.025 percent. Labor force and population is from the Census, as described in the Appendix. Gross Metropolitan Product is from the Bureau of Economic Analysis.

(ii) The regression line is $\text{Log GMP per capita} = 0.128 [0.015] * \text{Log population} + 9.46 [0.19]$. $R^2 = 0.44$ and $N = 100$.

Balland et al: Complex Economic Activities Concentrate in Large Cities (2020)

Research Questions:

Why do certain economic activities (e.g. technologies, industries, scientific fields) strongly concentrate in a few large cities?

What role does the complexity of an activity play in this spatial concentration? Has this concentration changed over time?

Data:

353 US Metropolitan Statistical Areas (MSAs)

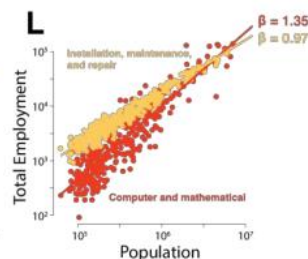
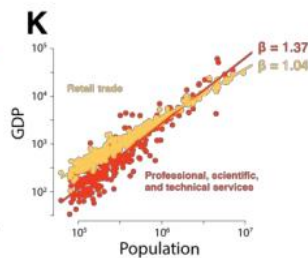
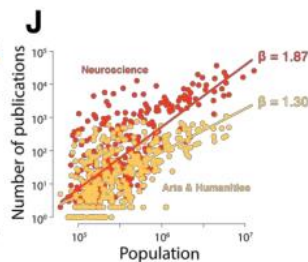
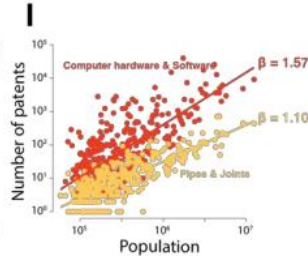
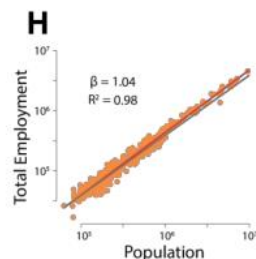
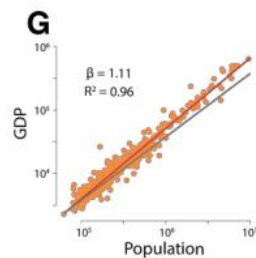
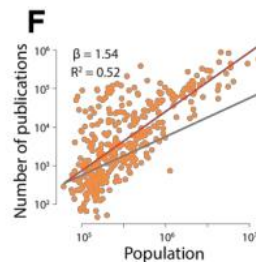
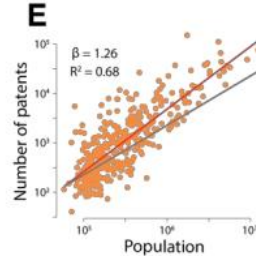
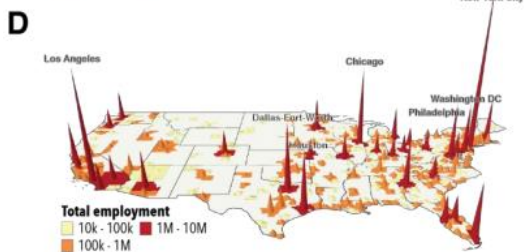
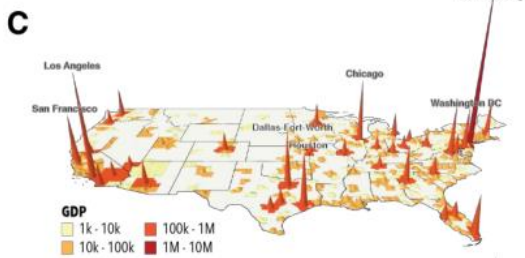
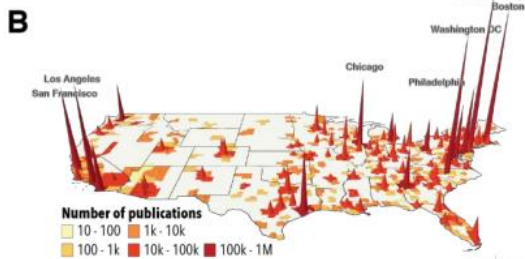
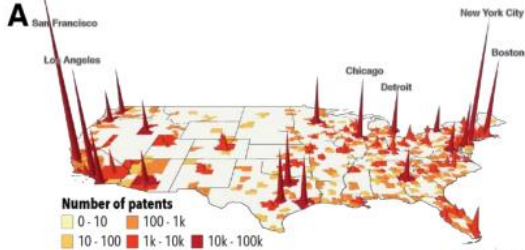
Indicators:

- Patents
- Scientific publications
- Industrial output
- Employment

Historical data (esp. patents) from ~1850 to present

Methods:

- Urban scaling analysis: Examines how activity levels scale with city size (superlinear, linear, sublinear relationships)
- Complexity measures: Based on collaboration size, technology novelty, knowledge combinations, etc.
- Time-series analysis: How concentration trends have evolved across decades for different complexity levels



Key Findings:

1. More complex activities are more spatially concentrated

- Activities with high complexity (e.g. biotech, neuroscience) are disproportionately located in a few major cities
- Complexity explains 40–80% of variance in spatial concentration (across industries, technologies, occupations, scientific fields)

2. Concentration increases over time

- The most complex technologies have become increasingly concentrated in the largest cities since the 1850s
- Even in the digital age, physical proximity remains critical for complex production & innovation

3. Reinforcing feedback loop

- Big cities attract complex activities → this fosters further urban agglomeration
- Smaller cities struggle to attract or retain such activities, leading to divergence

Takeaways:

- Innovation and complex production are not spreading evenly — they cluster in the largest cities
- Urban policy & regional development must consider the role of complexity and relatedness when promoting industrial growth
- Complexity is a key driver of spatial inequality in modern economies

IS INNOVATION (INCREASINGLY) CONCENTRATED IN LARGE CITIES? AN INTERNATIONAL COMPARISON

MICHAEL FRITSCH, MICHAEL WYRWICH

Is innovation becoming increasingly concentrated in large cities over time across different countries?

Data

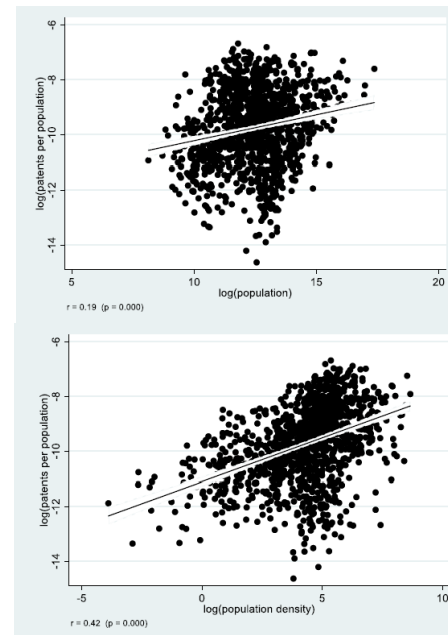
- OECD regional patent database (RegPat) covering 14 developed countries (2000-2015)
- Regional categories defined by the OECD functional urban areas

Methodology

- Compared patent concentration and innovation intensity: spatial distribution, patent-per-population ratios, and concentration indices (Herfindahl-Hirsch Index - HHI)
- Tested urban scaling and inventor productivity by regressing patent counts on regional population and inventor numbers, including checks for patent quality (citations, renewals, originality)

Findings

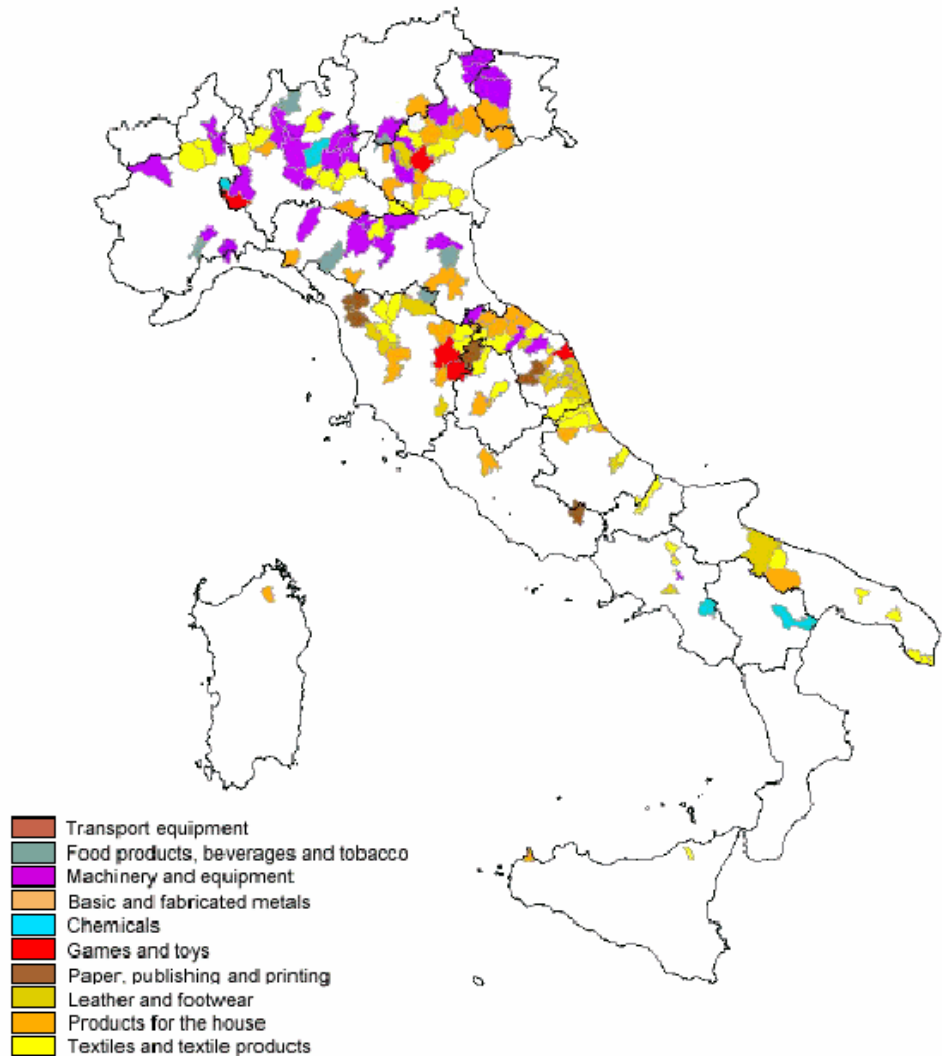
- The role of agglomeration economies is much less pronounced than many authors suggest
- Innovation is not increasingly concentrated in large cities, patents are more geographically concentrated than population but show no upward trend over time
- While big cities show higher patent counts, inventors are not more productive → innovation does not require large cities
- Centralizing R&D in metropolitan areas is inefficient, supporting regional and rural innovation is more balanced and effective



The specialization-diversity debate

Italian industrial districts

- SMEs (Small and Medium-sized Enterprises)
- Local traditions and strong cohesion
- Craft industry heritage
- Small town and its surroundings
- Business development organizations
- Supportive local governments



Specialization: export base

Location Quotient

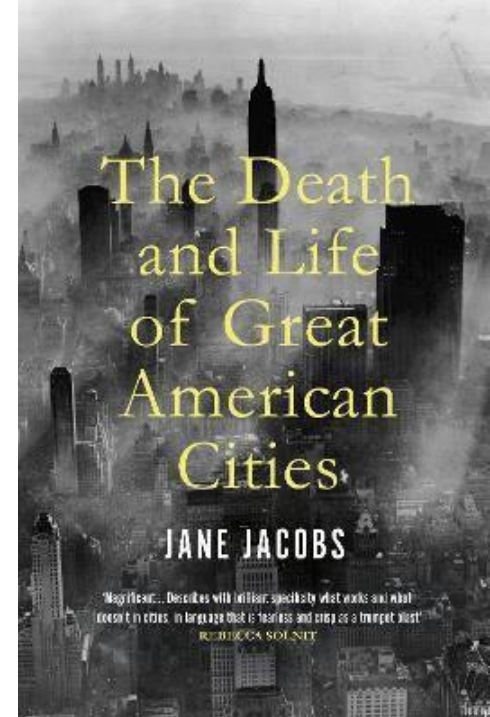
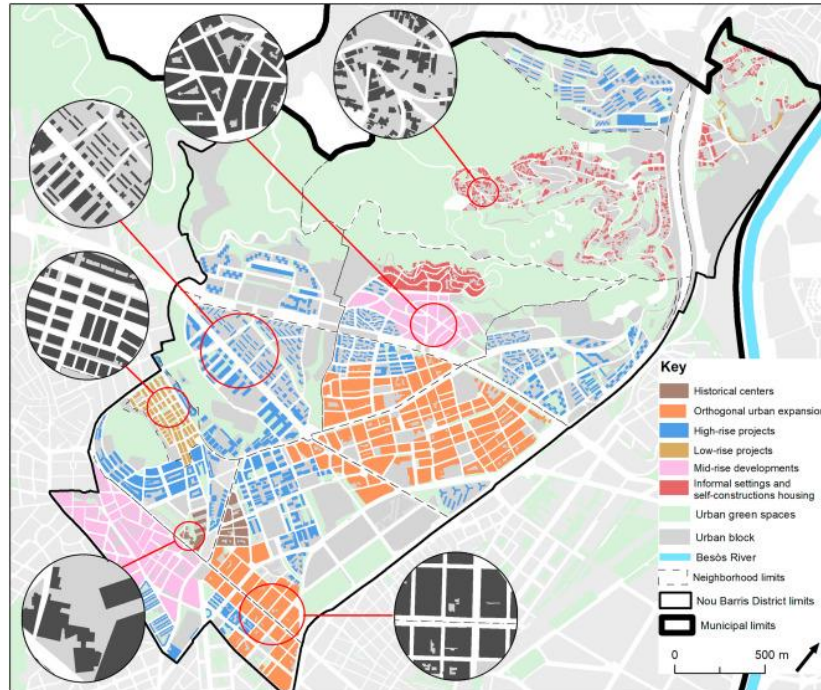
- Indicates the **degree of specialization** of a region in a given industry.
- Compares the **local share of employment** in the industry to the **national average**.
- Assumption: the **specialized product of the region is not only consumed locally**, thus the region participates in **interregional trade**.

$$LQ = \frac{\frac{E_{ir}}{E_r}}{\frac{E_{in}}{E_n}}$$



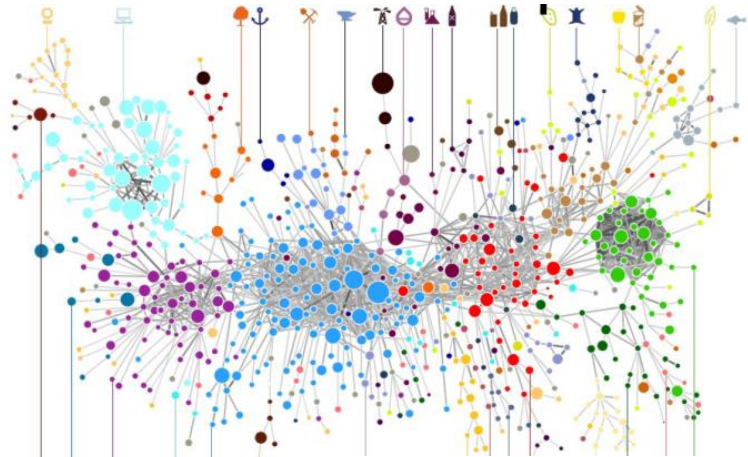
Diversity

Cities are arenas of interaction
Mixing in the built environment



Economic progress and urban success

- Social interaction: population density
- Learning in the city: industry structure
- Related knowledge is easier to learn but contains less novelty



Hidalgo et al. 2007 (Science)

Neffke et al. 2017

Labour flows in evolving regional economies

Zoltán Elekes



UMEÅ UNIVERSITET

(1) Background: labour flows and space

(2) Application: supporting regional economic diversification

(3) Application: workers escaping low-wage jobs

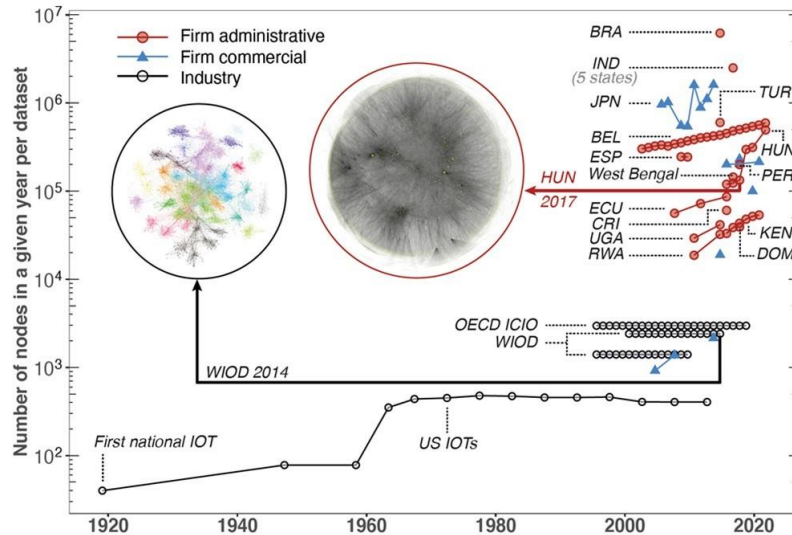
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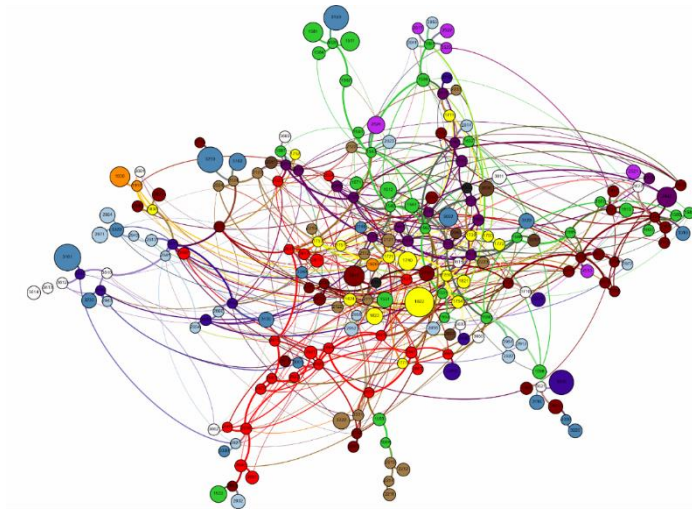
(3) Application: workers escaping low-wage jobs

(Spatial) economic networks

- Economic **networks** reflect a **division of labour** between the elements of an economy, and links reflect **solutions** to particular **co-ordination problems** inherent in the **production** of goods and services (Shutters et al. 2018 PLOS One).



Pichler et al. (2023) Science



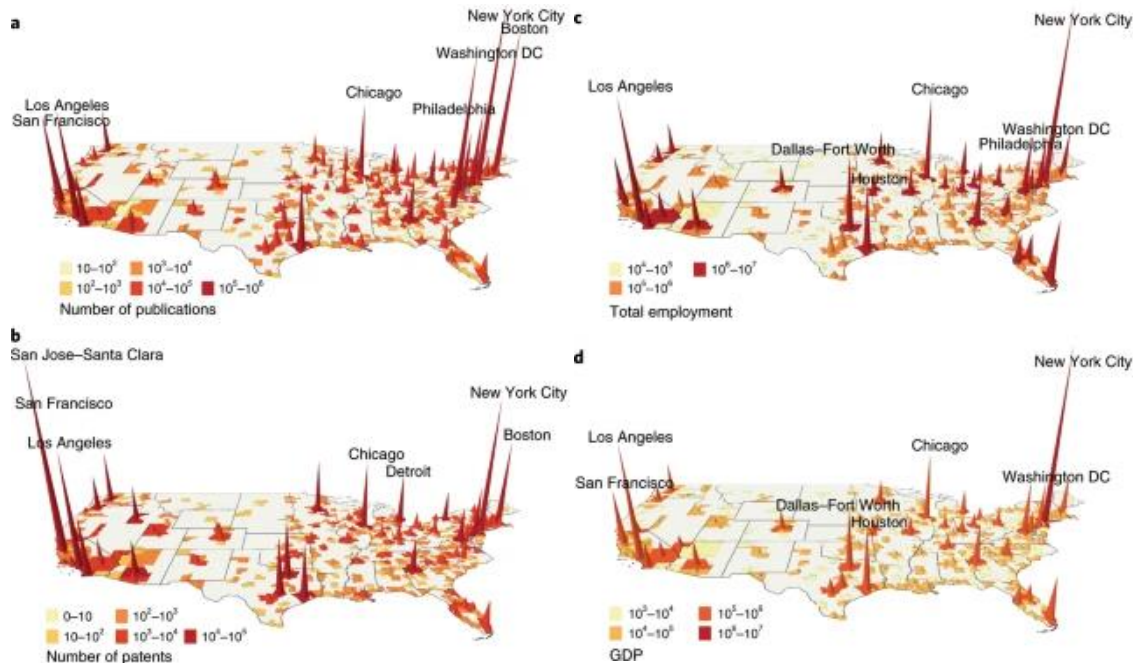
Elekes et al. (2019) RS

(Spatial) economic networks

- Uneven spatial distribution and clustering of economic activities.



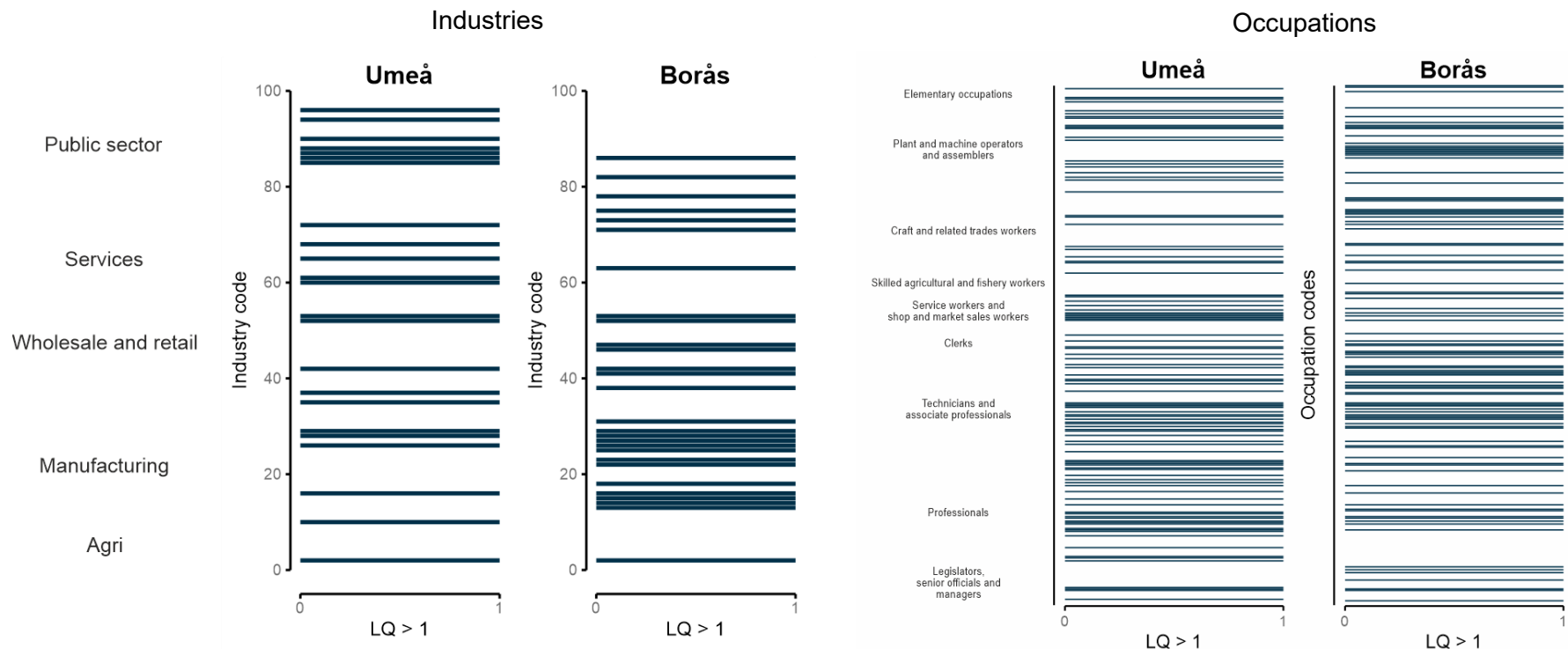
Florida (2005) The Atlantic Monthly



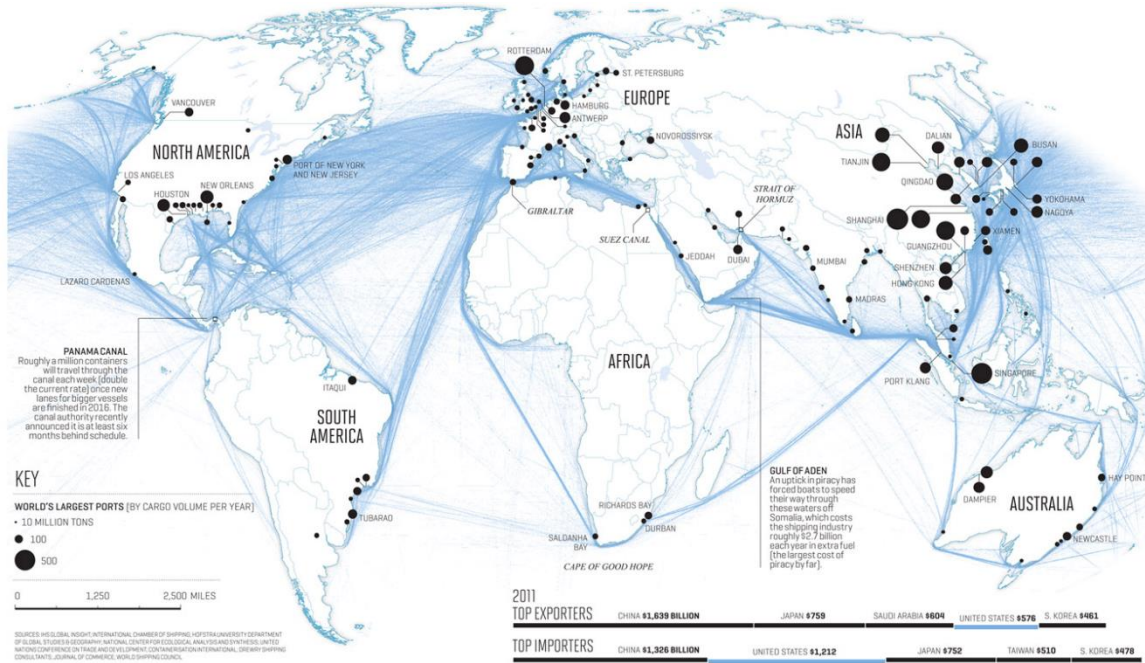
Balland et al. (2020) NatHB

(Spatial) economic networks

- Uneven spatial distribution and clustering of economic activities.
- We know that regional economies are **specialized and coherent**.



(Spatial) economic networks



- Uneven spatial distribution and clustering of economic activities.
- We know that regional economies are **specialized and coherent**.
- Clear division of labour within and between places.

Capability-based view of economic development

- Resource-based view of the region: sustained competitive advantage comes from **valuable, rare, imperfectly imitable** and **non-substitutable** resources (Neffke et al. 2018 EG, Frenken et al. 2023 CJRES).

Tools



Codes



Skills



- Capabilities are (to some extent) localised.

Capability-based view of economic development

- Resource-based view of the region: sustained competitive advantage comes from **valuable, rare, imperfectly imitable** and **non-substitutable** resources (Neffke et al. 2018 EG, Frenken et al. 2023 CJRES).

Tools



Codes



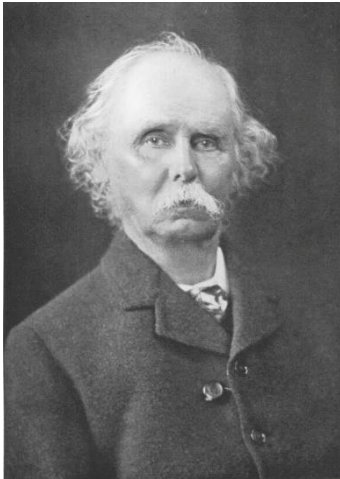
Skills



- Capabilities are (to some extent) localised.

Labour flows

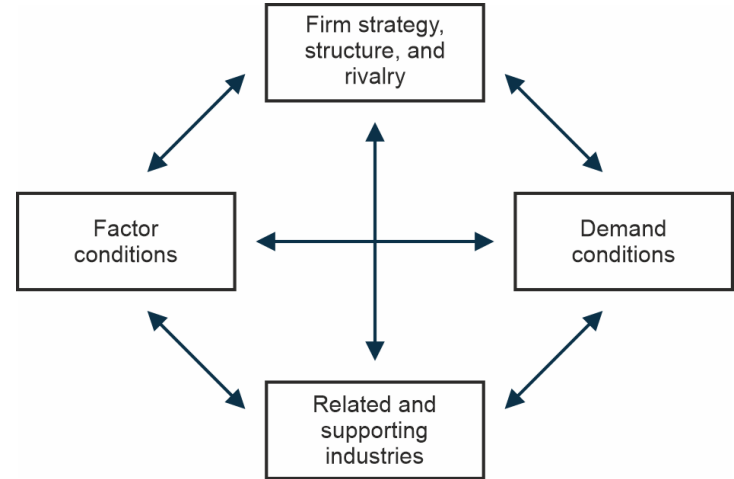
- **Marshallian channels:** (1) specialised local labour market; (2) specialised local suppliers; (3) local knowledge spillovers.
- **Porter's diamond:** factors of lasting competitive advantage -> regional clusters



Alfred Marshall

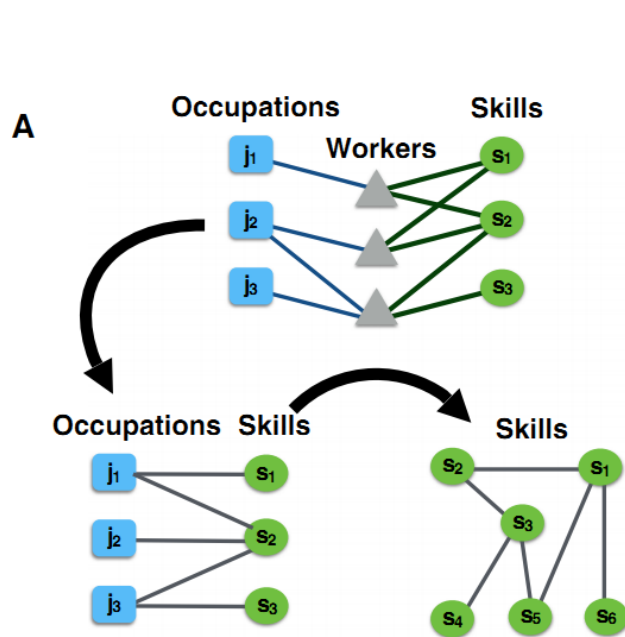


Michael Porter

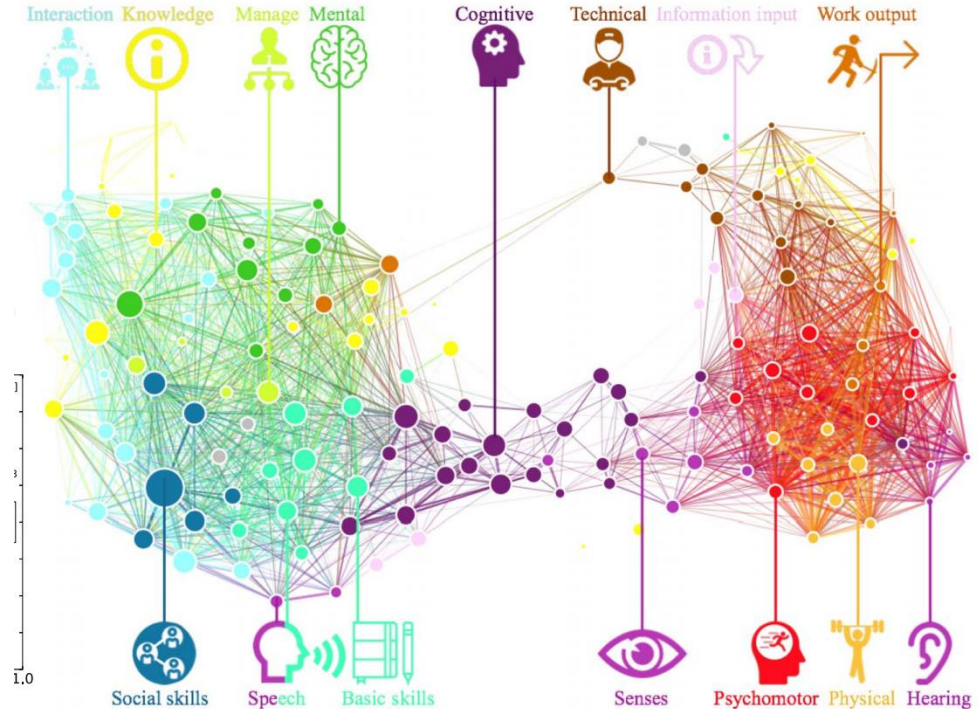


Skill-relatedness

- Economic activities are related through similarities in skill usage (e.g., Alabdulkareem et al. 2018 SciAdv, Hosseinioun et al. 2025 NatHB).

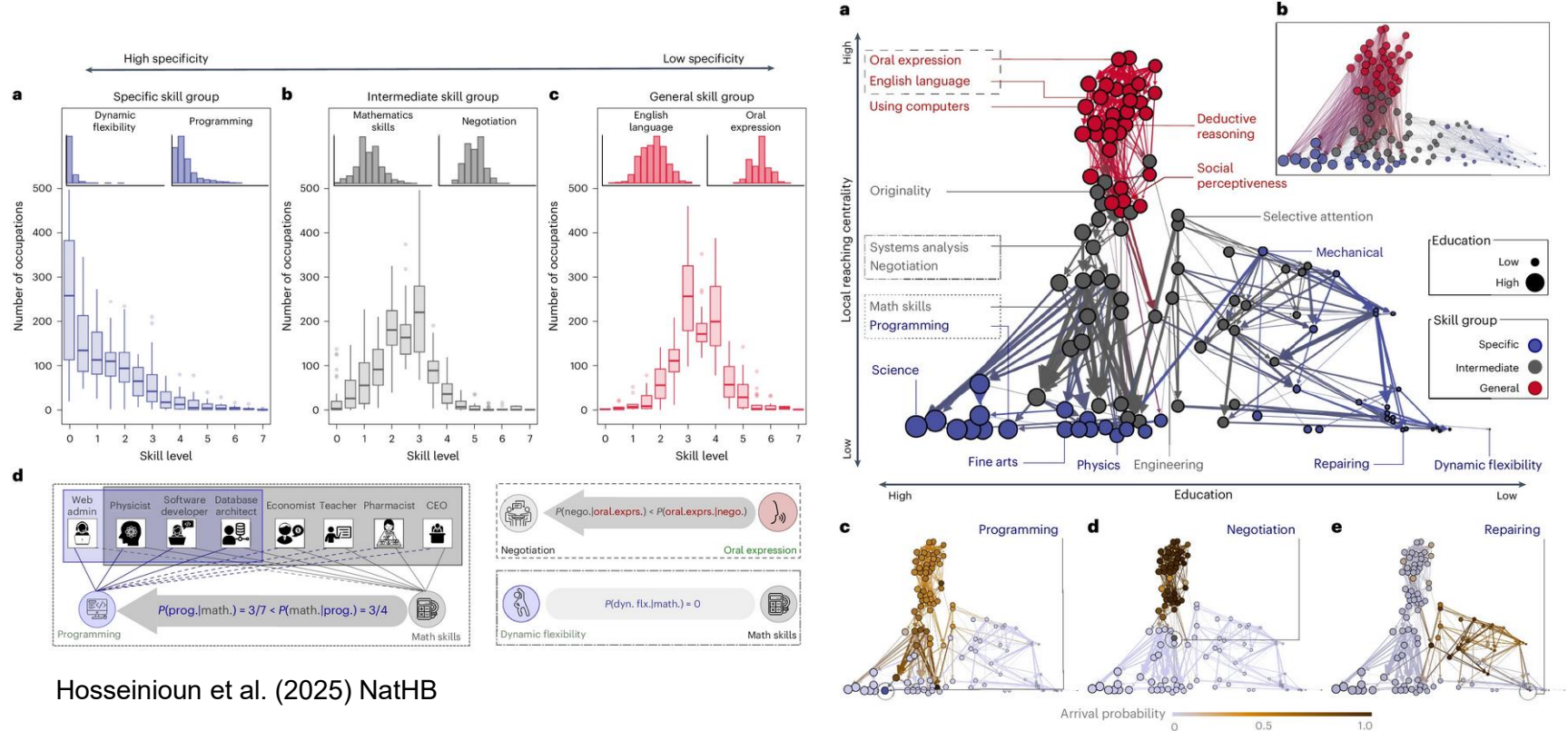


Alabdulkareem et al. (2025) SciAdv



Skill-relatedness

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Revealed skill-relatedness

- Direct observation of skills and skill demand seems to support the assumption about skill similarity behind revealed skill-relatedness (e.g., Neffke & Henning 2013 SMJ, Alabdulkareem et al. 2018 SciAdv, Henning et al. 2025 RSRS).

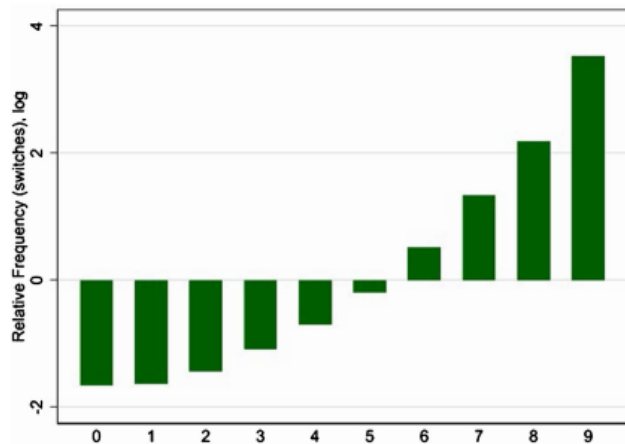
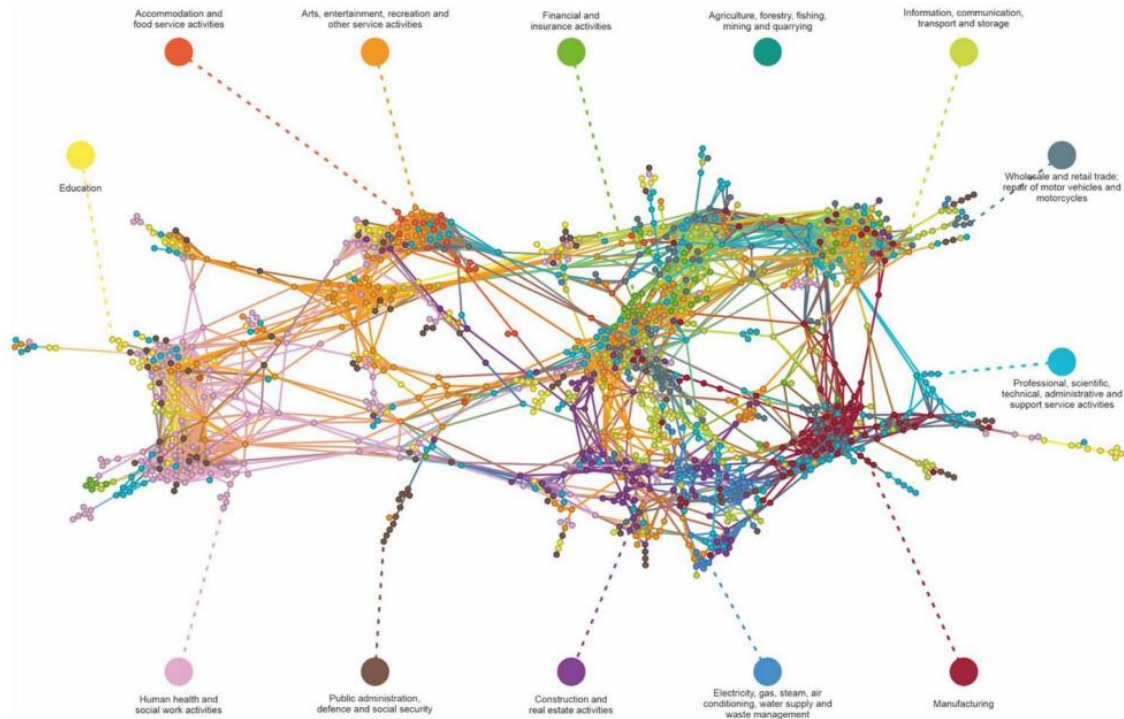


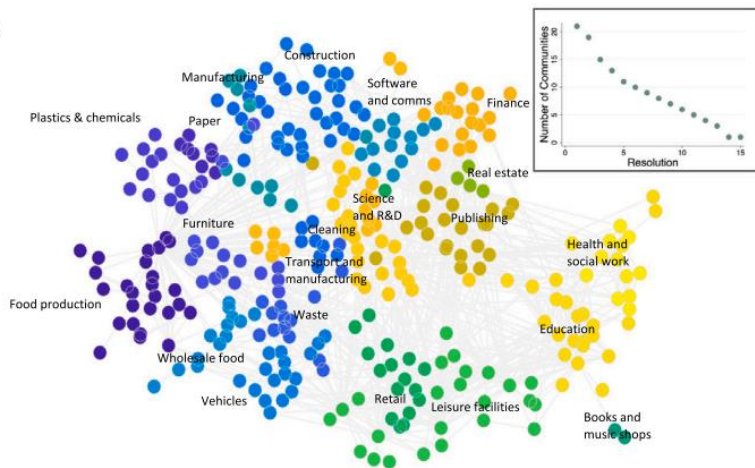
Figure 4. The relative frequency of job moves in Sweden between 2016 and 2020, per job-job relatedness strength (10 deciles).
Source: Our own elaboration on data provided by the Swedish Public Employment Service and Statistics Sweden.



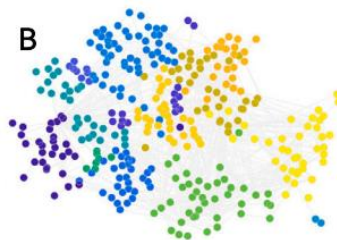
Henning et al. (2025) RSRS

Revealed skill-relatedness

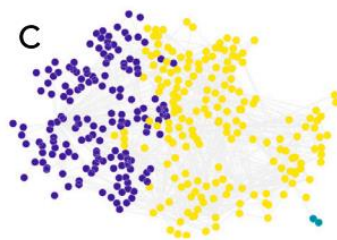
A



B

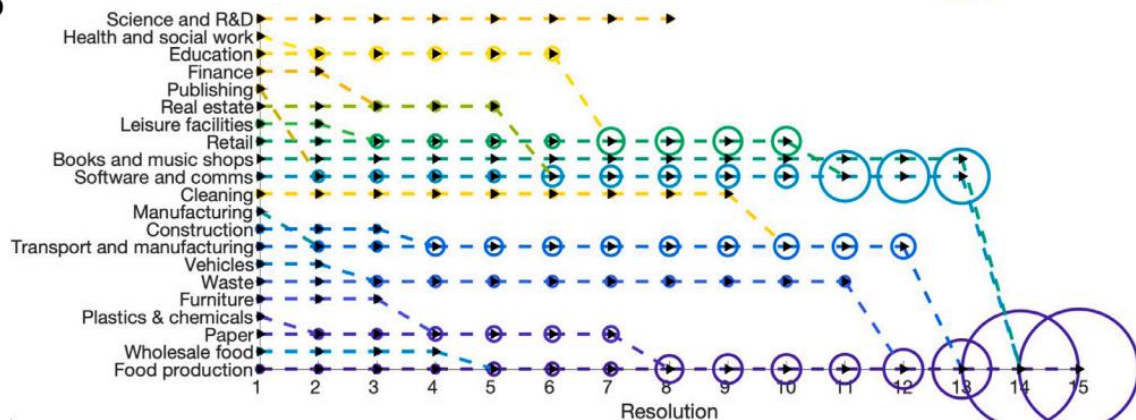


C



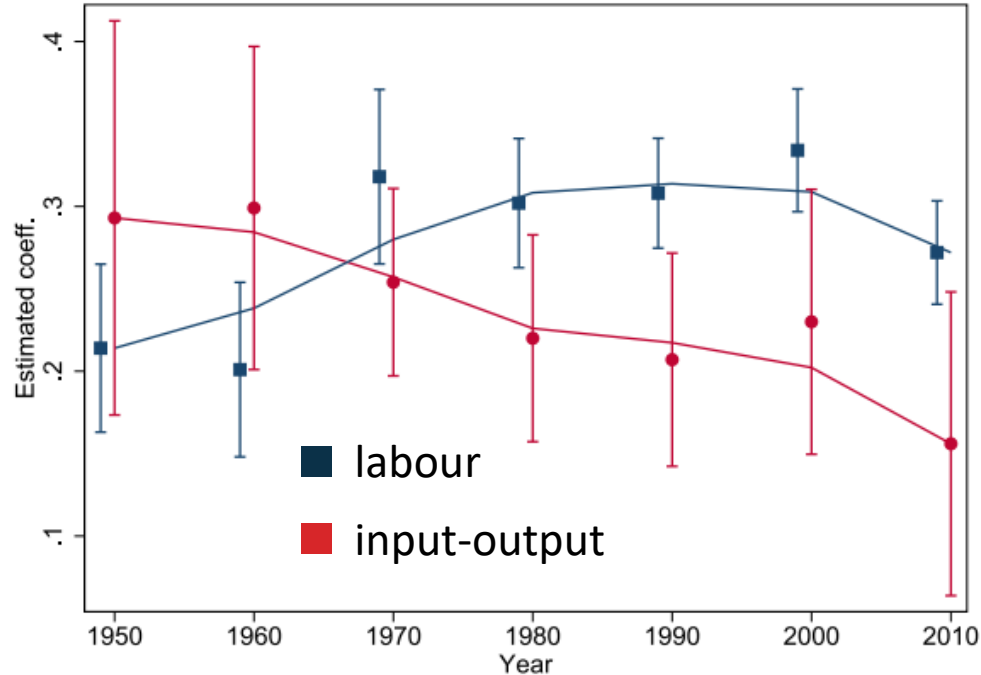
- Communities in industry spaces form “skill-basins” within which labour redeployment potential is higher (O’Clery & Kinsella 2022 RP), hence not only nearest neighbours matter.

D



Coagglomeration

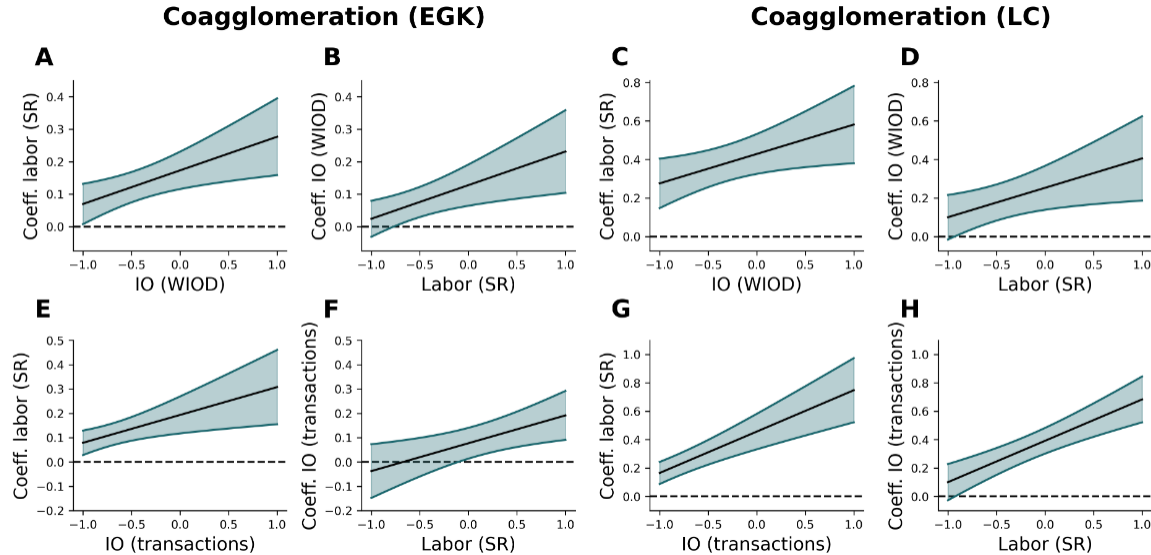
- Labour is a prominent Marshallian channel in driving the coagglomeration of industries (Ellison et al. 2010 AER).
- The importance the labour channel increased over time (Diodato et al. 2018 JUE).



Diodato et al. (2018) JUE

Coagglomeration

- Labour is a prominent Marshallian channel in driving the coagglomeration of industries (Ellison et al. 2010 AER).
- The importance the labour channel increased over time (Diodato et al. 2018 JUE).
- Labour flows carry knowledge (e.g., Boschma et al. 2009 RS, Eriksson 2011 EG, Lengyel & Eriksson 2017 JEG, Eriksson & Lengyel 2019 EG, Csáfordi et al. 2020 TJTT).

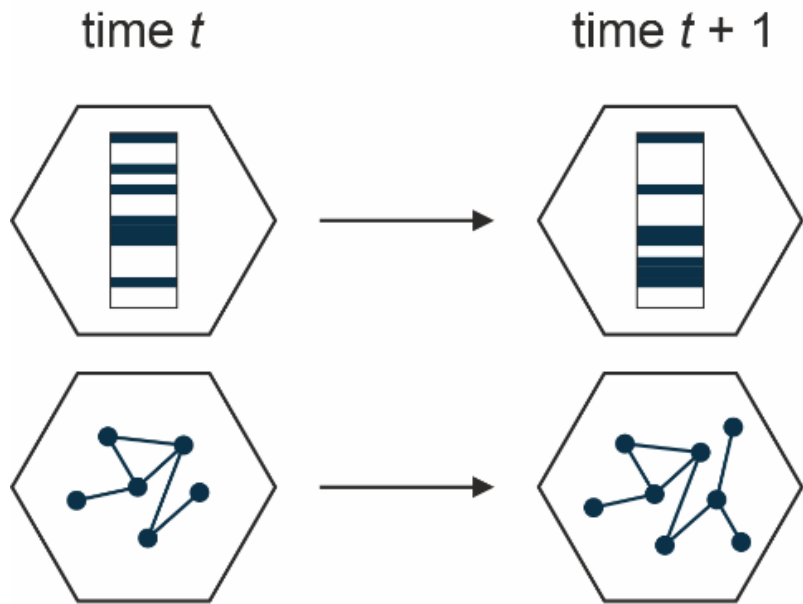


(1) Background: labour flows and space

(2) Application: supporting regional economic diversification

(3) Application: workers escaping low-wage jobs

<https://os-sweden.rec-group.se>



NETFLIX

for regions

(1) Background: labour flows and space

(2) Application: supporting regional economic diversification

(3) Application: workers escaping low-wage jobs

Regional diversification and labour market upgrading: local access to skill-related high-income jobs helps workers escaping low-wage employment

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This article investigates how the evolution of local labour market structure enables or constrains workers as regards escaping low-wage jobs. Drawing on the network-based approach of evolutionary economic geography, we employ a detailed individual-level panel dataset to construct skill-relatedness networks for 72 functional labour market regions in Sweden. Subsequent fixed-effect panel regressions indicate that increasing density of skill-related high-income jobs within a region is conducive to low-wage workers moving to better-paid jobs, hence facilitating labour market upgrading through diversification. While metropolitan regions offer a premium for this relationship, it also holds for smaller regions, and across various worker characteristics.

Keywords: skill-relatedness network, local labour market, low-wage workers, diversification and structural change, relatedness density

JEL Classifications: J21, J31, R11, R23

Introduction

Over a 20-year period, about 30% of local industries are replaced by new ones (Elekes et al., 2018), which is accompanied by even more drastic compositional changes in employment between sectors (Mecallie et al., 2006) and occupations (Hase-Wikman et al., 2022). In many advanced economies, structural change during the past few decades has brought about diminishing work opportunities in the middle of the wage distribution due to automation and offshoring of routine tasks (Goss et al., 2014; Gregory et al., 2022). The growing share of low-wage jobs, often with deteriorating working conditions, highlights the typical concern that these jobs offer limited career opportunities and might constitute a dead end for the workers involved (Axtell et al., 2004; Goss et al., 2019; Crayle and Kanelopoulos, 2018). This makes it imperative to under-

stand the individual and contextual factors underlying upward wage mobility. However, while numerous studies have explored the role of individual characteristics of low-wage workers in as well as the benefits of public policies for upward wage mobility (for example, Andersson et al., 2005; Lucifora et al., 2009; Cappellari and Jenkins, 2008; Pavlopoulos et al., 2012), studies on regional contextual factors facilitating upward wage mobility and escaping low-wage jobs are scarce.

This relative regional ignorance may bias our understanding of the prospects of escaping low-wage work. Because industries and occupations are unevenly distributed in space, county-level polarization in the wage distribution will necessarily play out differently on the underlying local labour markets, as exemplified for Sweden by Henning and Eriksson (2021). Additionally, the

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Regional diversification and labour market upgrading



Zoltán Elekes



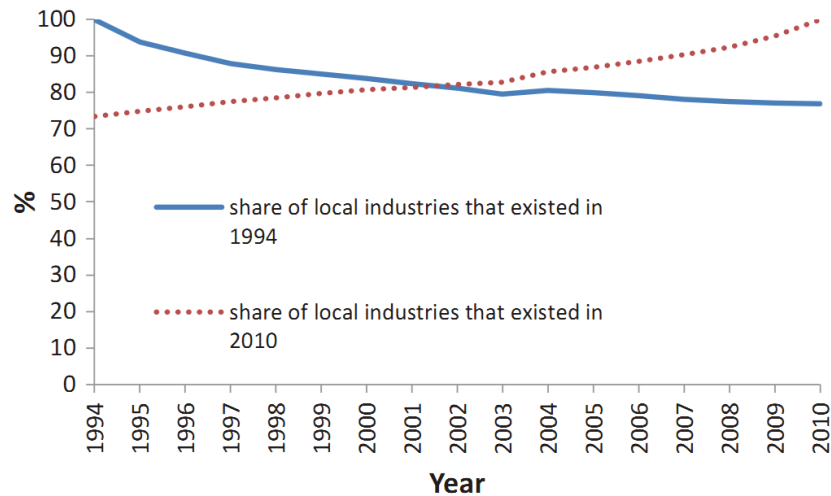
Anna Baranowska-Rataj



Rikard Eriksson

Motivation

- Relatively high turnover of economic activities, but a more stable coherence of economic activities.
- Related regional diversification.



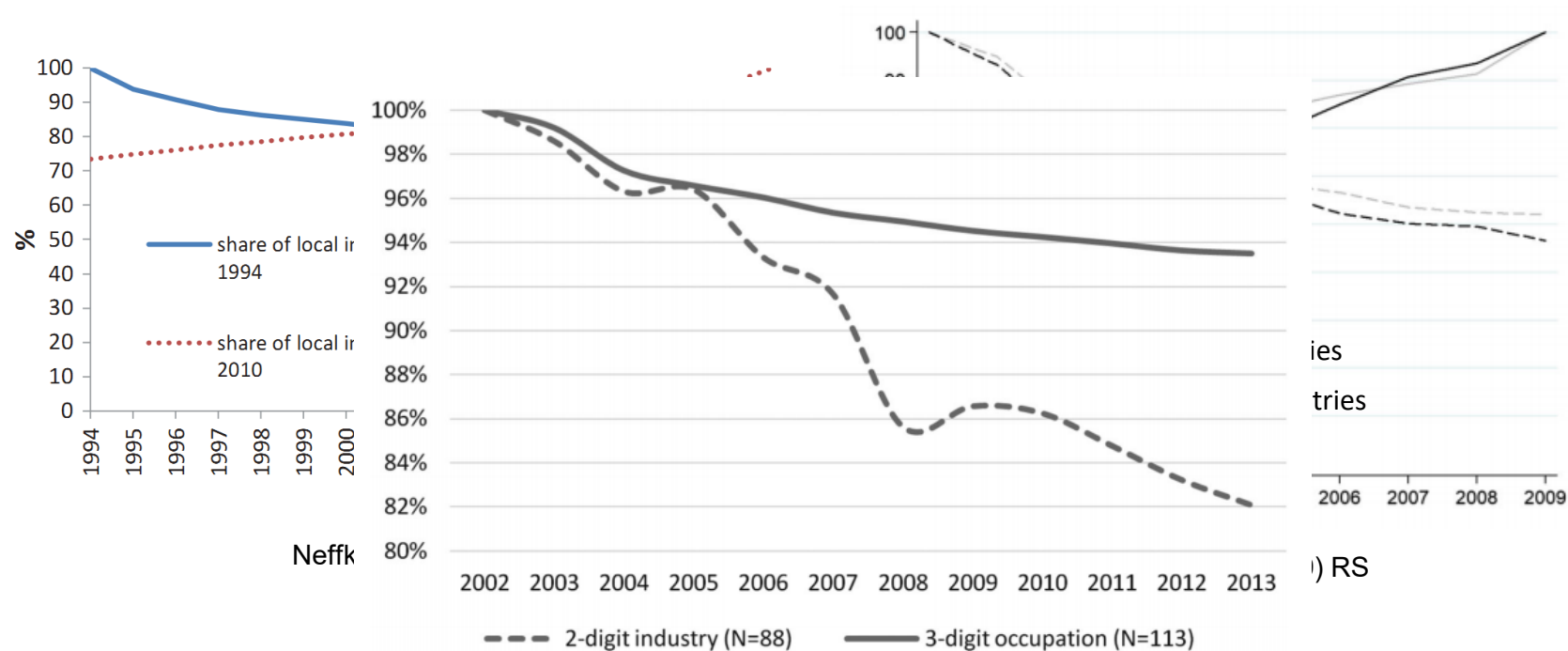
Neffke et al. (2018) EG



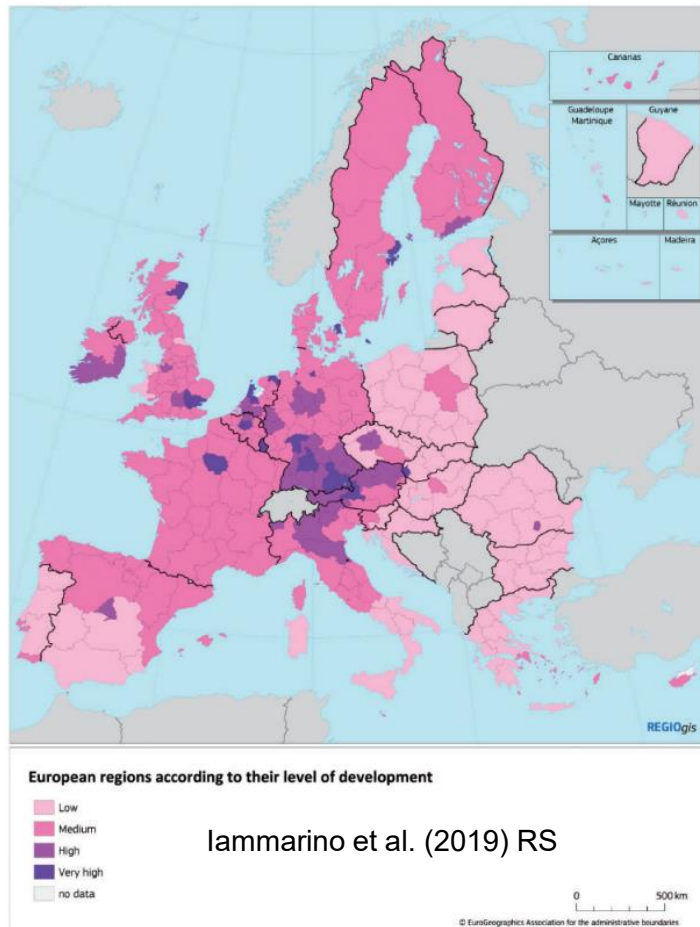
Elekes et al. (2019) RS

Motivation

- Relatively high turnover of economic activities, but a more stable coherence of economic activities.
- Related regional diversification.



Motivation



- Increasing inter- and intraregional inequalities (Iammarino et al. 2019 JEG).
- Social and political discontent in left-behind places (Rodríguez-Pose 2018 CJRES, Diemer et al. 2022 EG).

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The revenge of the places that don't matter (and what to do about it)

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Persistent poverty, economic decay and lack of opportunities are at the root of considerable discontent in declining and lagging-behind areas the world over. Poor development prospects and an increasing belief that these places have ‘no future’—an economic dynamism has been posited to be increasingly dependent on agglomeration economies—have led many of these so-called ‘places that don’t matter’ to revolt against the status quo. The revolt has come via an unexpected source: the ballot-box, in a wave of political populism with strong territorial, rather than social foundations. I will argue that the populist wave is challenging the sources of existing well-being in both the less-dynamic and the more prosperous areas and that better, rather than more, place-sensitive territorial development policies are needed in order to find a solution to the problem. Place-sensitive development policies need, however, to stay clear of the well-known, incorrect support and big investment projects of past development strategies if they are to be successful and focus on tapping into untapped potential and on providing opportunities to those people living in the places that ‘don’t matter’.

Keywords: economic development, territorial inequality, regions, cities, populism, place-sensitive policies.
JEL Classification: P63, O18, R11, R12.

From hectoring to revenge

On 16 October 2008, Tim Leung, then a Reader in Economic History at the London School of Economics, stood in Liverpool’s Cathedral in front of a crowd of somewhat bemused and worried Liverpoolians to essentially deliver the message that Liverpool’s time was past; that the place that they lived in and loved no longer mattered. Cities and counties in the North of England had ‘slipped back relative to both the national average and Britain’s most successful towns’ and ‘regeneration policy [had] failed to regenerate towns’ (Leung, 2008). Despite

massive public expenditure in trying to promote development in the North, the results were dismal: the economic gap between a prosperous South of England and a declining North had continued to widen regardless of the efforts of successive governments (Martin et al., 2015). Development policies were not working and there was a need to rethink development strategies for lagging and declining areas in the UK, as the fate linked to geography could not be bucked (Leung, 2008).

The proposed solution was relatively simple: first, focus on the parts of the country that

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The Regional Development Trap in Europe

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abstract The concept of *regional development trap* refers to regions that face significant structural challenges in retrieving past dynamism or improving prosperity for their residents. This article introduces and measures the concept of the regional development trap for regions in Europe. The concept draws inspiration from the *middle-income trap* in international development theory but widens it to shed light on traps in higher-income countries and at the regional scale. We propose indicators—involving the economic, productivity, and employment performance of regions relative to themselves in the immediate past, and to other regions in their respective countries and elsewhere in Europe—to identify regions either in a development trap or at significant near-term risk of falling into it. Regions facing development traps generate economic, social, and political risks at the national scale but also for Europe as a whole.

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Research design

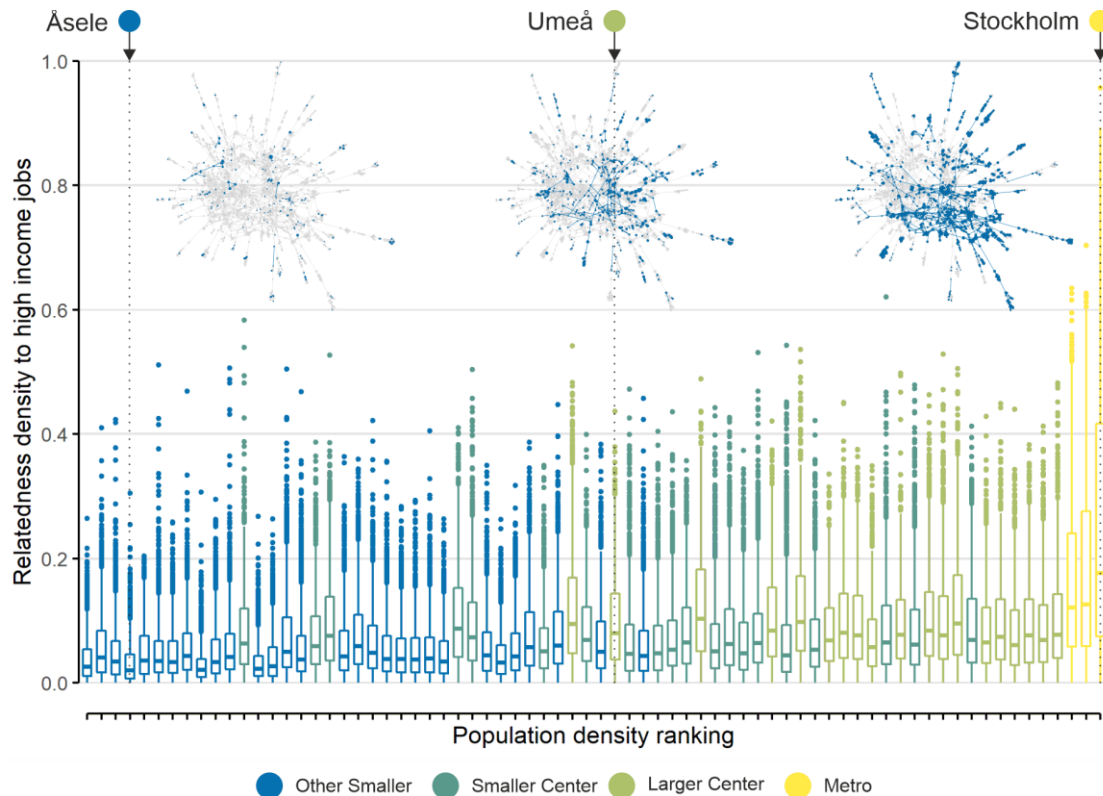
- Data: matched employer-employee register data on the universe of workers in Sweden (2002–2012).
- A “**job**” is an industry-occupation combination (c.f., Henning & Eriksson 2021 CJRES).
- Based on the median income distribution of jobs in 2002, a high-wage job belongs to the top two quintiles (~ 60% median income).
- **Revealed skill relatedness**: significant labour flows between jobs from 2002-2005.
- **Relatedness density**: To what extent skill-related jobs (likely channels of mobility) are concentrated in a local labour market for any given job
- Specifically, what percent of total relatedness (total weighted degree) connects a job to jobs showing relative specialization in a region.
- **High-income relatedness density**: considering only relatedness to those jobs, that are in the top 2 quintiles of the income distribution.

$$RD_{i,r,t} = \frac{\sum_i SR_{ij} I(LQ_{j,r,t})}{\sum_i SR_{ij}}$$

$$HI. RD_{i,r,t} = \frac{\sum_i SR_{ij} I(LQ_{j,r,t}) I(ICAT_j)}{\sum_i SR_{ij}}$$

Research design

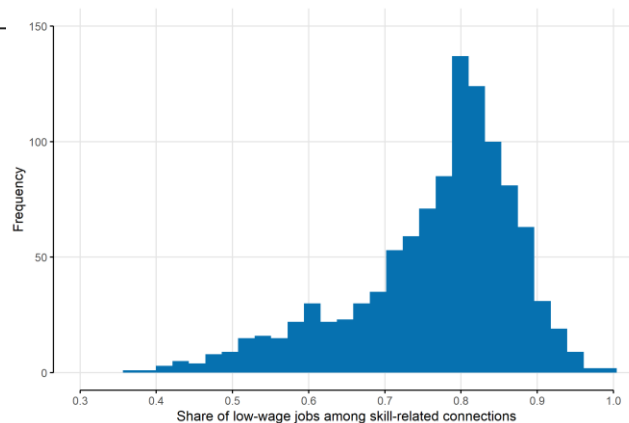
- Network assortative to median-income of jobs (corr.: 0.46).



Research design

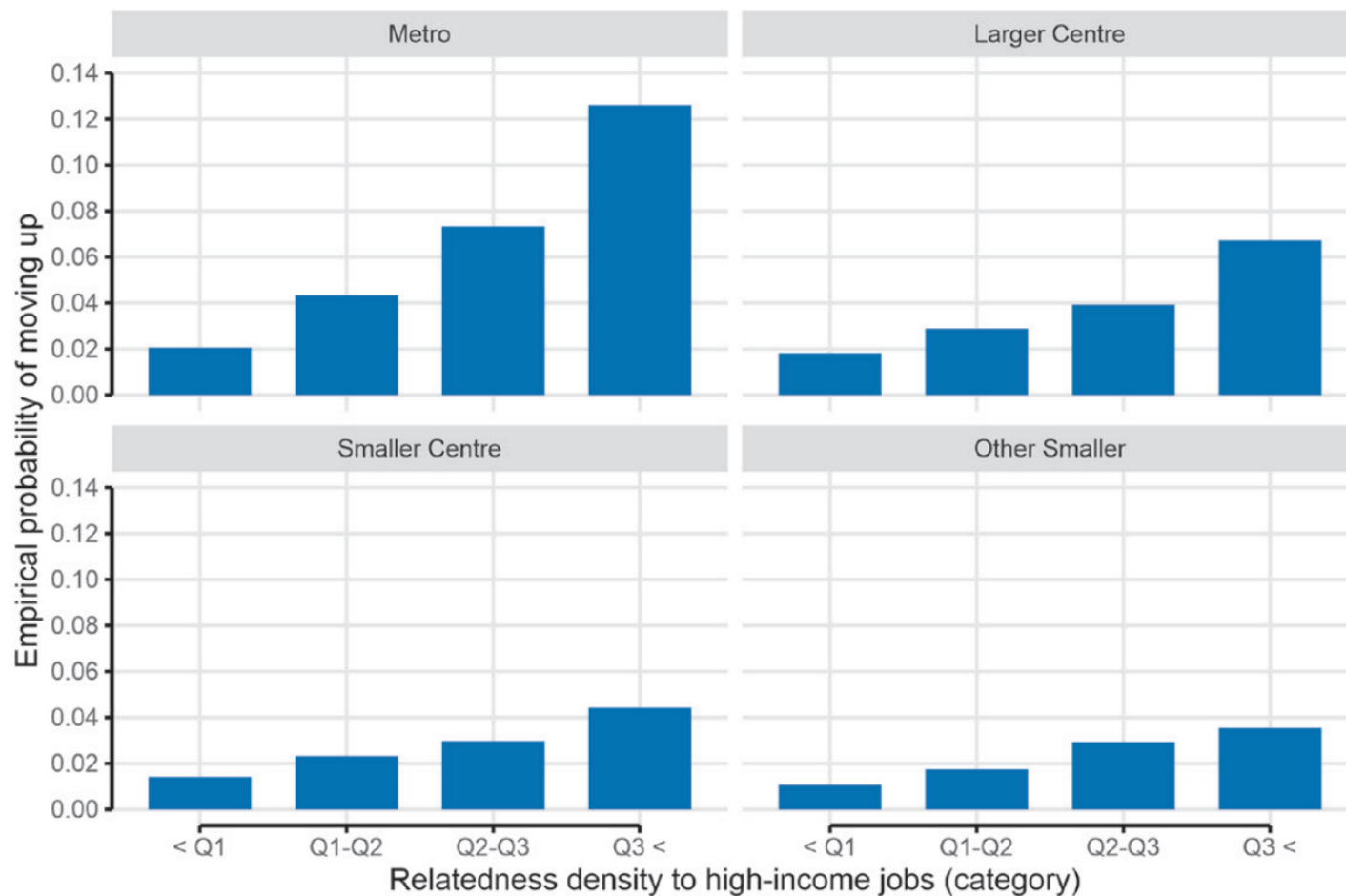
The share of low-wage connections among the most frequent low-wage jobs.

Job (industry × occupation)	<i>N</i> (person-year)	Low-wage share (%)
87.513: Personal care and related workers in Residential care activities	1300947	0.94
47.522: Shop and stall salespersons and demonstrators in Retail trade, except of motor vehicles and motorcycles	1109737	0.93
86.513: Personal care and related workers in Human health activities	719051	0.84
85.513: Personal care and related workers in Education	589246	0.94
85.331: Pre-primary education teaching associate professionals in Education	587272	0.86
85.233: Primary education teaching professionals in Education	587172	0.74
49.832: Motor-vehicle drivers in Land transport and transport via pipelines	575916	0.78
86.323: Nursing associate professionals in Human health activities	416377	0.65
88.513: Personal care and related workers in Social work activities without accommodation	313669	0.89
41.712: Building frame and related trades workers in Constructions of buildings	230843	0.79



Majority of links to other low-wage jobs (horizontal links)
while few offer upward career channels

Results



Results

1. Increasing density of related jobs associated with an increased likelihood of upward mobility (M2).

Table 1. Fixed-effect linear probability models on the within-regional career mobility of low-wage workers 2006–2012.

LHS (sample)	(1)	(2)	(3)	(4)	(5)	(6)
	Into top 2 (all LW)	Into top 2 (all LW)	Into top 2 (all LW)	Into top 2 (all LW)	Into top 3 (LW1-2)	Income change (all LW)
RD_{t-1}		0.003*** (0.001)				
$HI.RD_{t-1}$			0.070*** (0.003)	0.071*** (0.003)	0.121*** (0.005)	0.262*** (0.004)
AGE_{t-1}	0.007*** (0.000)	0.007*** (0.000)	0.007*** (0.000)	0.005*** (0.000)	0.010*** (0.000)	0.014*** (0.000)
$\ln INCOME_{t-1}$	0.009*** (0.000)	0.009*** (0.000)	0.009*** (0.000)	0.009*** (0.000)	0.015*** (0.000)	-0.783*** (0.001)
$PCT.HINCJOB_{t-1}$	0.100*** (0.011)	0.102*** (0.011)	0.098*** (0.011)	0.017 (0.011)	0.222*** (0.017)	0.012 (0.020)
$\ln POPDENS_{t-1}$	0.005*** (0.000)	0.004*** (0.000)	0.003*** (0.000)	0.190*** (0.004)	0.004*** (0.001)	0.014*** (0.001)
$PCT.WORKAGE_{t-1}$	0.277*** (0.011)	0.275*** (0.011)	0.261*** (0.011)	0.399*** (0.014)	0.396*** (0.017)	0.495*** (0.020)
$AVG.WPSIZE_{t-1}$	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.012*** (0.001)	-0.003*** (0.001)	-0.007*** (0.001)
$\ln JOBSIZE_{t-1}$	-0.004*** (0.000)	-0.004*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.007*** (0.000)	-0.000 (0.000)
Constant	-0.440*** (0.005)	-0.441*** (0.005)	-0.437*** (0.005)	-1.261*** (0.017)	-0.681*** (0.008)	3.554*** (0.011)
Worker FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Region FE	No	No	No	Yes	No	No
N (worker-year)	13,759,534	13,759,534	13,759,534	13,759,534	9,124,751	13,759,534
n (worker)	3,199,126	3,199,126	3,199,126	3,199,126	2,296,099	3,199,126
Within R ²	0.011	0.011	0.012	0.012	0.017	0.432

Notes: standard errors in parentheses; *p < 0.1, **p < 0.05, ***p < 0.01.

Results

1. Increasing density of related jobs associated with an increased likelihood of upward mobility (M2).
2. The likelihood of leaving a low-income job for a higher-income job is higher when the region upgrades into related high-income jobs (M3-4).

Table 1. Fixed-effect linear probability models on the within-regional career mobility of low-wage workers 2006–2012.

LHS (sample)	(1)	(2)	(3)	(4)	(5)	(6)
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Results

1. Increasing density of related jobs associated with an increased likelihood of upward mobility (M2).
2. The likelihood of leaving a low-income job for a higher-income job is higher when the region upgrades into related high-income jobs (M3-4).
3. Robust over different definitions of low-wage (e.g., M5-6), relatedness density, co-mobility, moving to new workplace, adverse outcomes (downward mobility, moving to other regions).

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ln JOBSIZE _{t-1}	-0.004*** (0.000)	-0.004*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.007*** (0.000)	-0.000 (0.000)
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Within R ²	0.011	0.011	0.012	0.012	0.017	0.432

Notes: standard errors in parentheses; *p < 0.1, **p < 0.05, ***p < 0.01.



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Assignment: Research Proposal

- Prepared in groups of 2
- Content (*example*)
 - research problem: *Inequalities are growing across cities.*
 - question: *How are spatial inequalities related to urban networks?*
 - description of data collection: *We use Facebook data across US cities.*
 - methodology: *Club convergence estimation on Facebook network communities.*
 - expected outcomes: *Income converge within network communities but diverge between them.*
 - impact: *stakeholders (US ministries, federal agencies), venues of distribution of results to reach professional and public audience*
 - references.
- The written proposal must be prepared in pdf format (font size 12, line spacing 1.5, 2.5 margins, times new roman font, page numbering) and then must be presented in the class. Coverage: 5-7 pages, plus references. The 7-page limit is strict!
- Deadline for paper submission: November 16. Presentation: November 18 (week 8). Presentation cannot be more than 5 minutes each!